

What More from a Smartphone?

Today, a smartphone can browse the Web, play HD video, run the latest applications, handle interactive 3D gaming and have inbuilt e-mail, chat and social networking solutions. How does this palm-sized device manage all this?



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There is a lot more to smartphones than just the operating system (OS) and an app count. However refined OS a phone has, the performance and, more importantly, the productivity the user gets also depends on the number-crunching power the hardware packs within. The race to improve performance is not just driven by megahertz or umpteen central processing unit (CPU) cores anymore.

Before getting into the nitty-gritty, it's important to know that smartphones and tablets use a system-on-chip (SoC). SoC is the equivalent of

a computer motherboard including central processing unit, graphics processing unit and memory—on a single chip. There is no way to predict the performance of a smartphone by simply checking out its processor. SoCs with even the same processors have greatly varying performance. This could be due to different configurations (clock speed, GPUs, etc). There is also the chance of some vendors cheating the benchmark test by tuning their product for this particular test often at the expense of worse performance on actual work load.

However, a smartphone with a high-end SoC does have some

expectations to live up to. In this article, we provide benchmarks for most of the popular SoCs so that you can compare these. Remember though, that the benchmark results indicated here may tell a significantly different story from real-world performance, especially for multi-core SoCs. This is true especially when the OS in question has been optimised for those multiple cores.

SoC processor cores

Often, SoCs use processors and design from British company ARM Holdings plc. One of the primary reasons why Advanced RISC Machines (ARM) processors came to be so widely

used in SoCs for mobile devices and portables is their low electric power consumption. For CPUs licensed from ARM Holdings plc, the corresponding GPUs are licensed from Imagination Technologies, which is known for its PowerVR graphics cards. PowerVR graphics processor designs are licensed to many SoC makers including Samsung, Apple, Texas Instruments, Intel, NEC, NXP and Freescale.

Another popular core is the Snapdragon core *aka* Scorpion. It is designed and built by Qualcomm using the ARM v7 instruction set. Snapdragon is considered to perform better for multimedia-related single-instruction, multiple-data (SIMD) operations. The graphics processors in Qualcomm's SoCs are usually Adreno flavoured processors from Imageon—a subsidiary of Qualcomm and descendant of ATI.

Another core is the Intel Atom CPU for SoCs featured on the mobile Internet devices (MIDs). Intel SoCs are also paired with PowerVR SGX GPUs. The last release was the Moorestown platform with a 45nm Atom CPU.

How good a processor core is for multi-CPU designs depends on:

1. Performance density in the form of maximum aggregate performance per watt or per square millimetre
2. Inter-processor communications that minimise inefficiencies in the partitioning boundaries

Benchmark tests

There are various software which are used on the complete device to test its processor's performance:

BrowserMark benchmark. The BrowserMark benchmark test measures a browser's performance in JavaScript and HTML rendering (refer